MLOPS

# Model Monitoring & Drift Visual Guide

Catching decay before users do:
data drift, concept drift, PSI/KS, retraining triggers.

**ANIMATED WALKTHROUGH**
Step through pages to play the animation

STEP 1/5

# Two Kinds of Drift

Inputs shift, or the world changes meaning.

**Data drift**

customers get younger, sensors recalibrated.

P(X) changes:

**Concept drift**

what 'fraud' means evolves over time.

P(y|X) changes:

Data drift is detectable without labels. Concept drift needs ground truth — the harder, more dangerous kind.

STEP 2/5

# Detecting Drift

Compare live distributions to training.

**PSI**
Population Stability Index: binned distance between distributions.

**KS test**
Kolmogorov-Smirnov: max gap between CDFs; p-value for "same?".

PSI > 0.25 or KS p < 0.05 on key features = investigate. Track per-feature, not just globally.

Monitor the features the model actually depends on (top SHAP features) — drift elsewhere may not matter.

STEP 3/5

# Label-Free Monitoring

Watch the model when ground truth is slow.

- Prediction distribution: a fraud model suddenly approving 40% more transactions is screaming.
- Confidence/entropy: rising uncertainty often precedes accuracy drops.
- Input schema checks: new categories, null spikes, unit changes — catch pipeline breaks first.
- Shadow labels: sample predictions for human review weekly to estimate true accuracy cheaply.

Labels can lag by weeks (did the loan default?). Label-free signals are your early warning system.

STEP 4/5

# Retraining Triggers

Decide in advance when to retrain.

### Schedule

Retrain weekly/monthly — simple, predictable, sometimes wasteful.

### Drift threshold

PSI breach on key features fires a retraining pipeline automatically.

### Performance decay

Live accuracy / business KPI drops below the agreed floor.

### Data event

New product launch, regulation change, or upstream schema migration.

STEP 5/5

# The Drift Response Playbook

From alert to action.

- 1. Triage: is it a pipeline bug (null spike) or real drift? Check data quality first.
- 2. Scope: which features drifted, and does the model depend on them?
- 3. Decide: retrain on recent data, reweight, or fix the upstream source.
- 4. Validate the challenger against the champion on recent labeled data before promoting.
- 5. Post-mortem: log the drift event — patterns across incidents reveal systemic issues.

STEP ✓/✓

# Key Takeaways

## Model Monitoring & Drift — Remember

- Data drift = inputs shift (detectable without labels); concept drift = meaning shifts (needs labels).
- PSI and KS tests compare live vs training distributions per feature.
- Monitor prediction distributions and confidence when labels lag.
- Set retraining triggers in advance: schedule, drift threshold, or performance floor.
- Triage pipeline bugs first; validate challengers before promoting.

**Keep exploring**

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev